IDSC 2026 PITCH DECK

Team VZ

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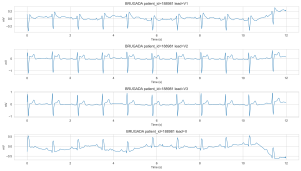


We do not just optimize accuracy. We optimize for fewer missed high-risk patients.

The Healthcare Problem

Why this matters

- Brugada syndrome can lead to sudden cardiac death, but its ECG pattern can be subtle.
- In screening, a false negative is more dangerous than an extra false alarm.
- Many AI studies chase accuracy, even when the real clinical question is: **who are we still missing?**

Our pitch

Use ordinary 12-lead ECGs to build a clinically honest screening aid that helps specialists miss fewer dangerous cases.

356

binary ECG patients after excluding atypical label-2 cases

69

Brugada cases in the full benchmark cohort

11

ECG-only models compared on one shared patient split

Our Mathematical and Computational Solution

Pipeline

1. Start from raw WFDB 12-lead ECGs.
2. Build median-beat and engineered-feature representations.
3. Train 11 complementary models on the same patient-level split.
4. Tune thresholds on validation data, then test once.

Clinical design choice

The primary benchmark is **ECG-only**. Metadata stays in a separate ablation to avoid shortcut learning.

Safety-aware ranking

$$\text{Composite} = 0.40P + 0.35S + 0.15C + 0.05E + 0.05T$$

- P: performance
- S: safety
- C: consistency
- E: efficiency
- T: stability evidence

Key idea

We do not let raw leaderboard score hide a poor false-negative profile.

Why 11 Models Matter for Healthcare Hope

Feature-based models

- **XGBoost**: strong nonlinear baseline on engineered ECG biomarkers.
- **Random Forest**: robust ensemble for noisy and variable patient signals.
- **SVM-RBF**: captures subtle curved decision boundaries in borderline ECGs.
- **AdaBoost**: focuses learning on the hardest-to-classify cases.

End-to-end deep models

- **1D CNN Median Beat**: learns local waveform cues directly from ECG morphology.
- **ResNet-1D Median Beat**: goes deeper without unstable training, improving safety-oriented detection.
- **CNN + BiGRU Median Beat**: combines local patterns with sequential beat context.
- **Transformer Encoder Median Beat**: models wider lead-to-lead and time interactions.

Transfer and advanced models

- **Transfer Learning (In-domain DAE to Classifier)**: reuses ECG structure learned before classification, helping when labeled data is limited.
- **VICReg (Self-supervised)**: learns useful signal representations without needing labels on every step.
- **Echo State Network (ESN)**: provides a lightweight dynamic sequence baseline for efficient screening.

Why this brings hope: we did not trust one lucky algorithm. We tested 11 mathematical views of the same ECG so clinicians get a more credible, safety-aware, and practical screening aid.

What the Benchmark Found

Best single recommendation

Transfer Learning (In-domain DAE to Classifier)

- PR-AUC: 0.676, F1: 0.692
- Sensitivity: 0.643, Specificity: 0.948
- False Negatives: 5
- Selected by the balanced multi-aspect ranking

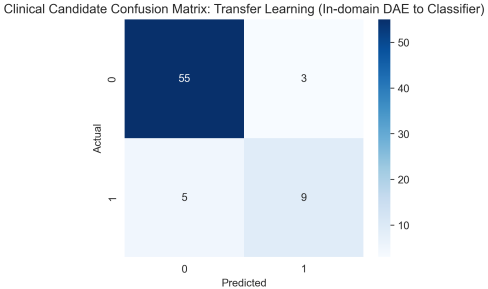
Best screening-safety story

Transfer Learning (In-domain DAE to Classifier)

- Sensitivity: 0.643, Specificity: 0.948
- False Negatives: 5, False Positives: 3
- The same model also won the clinical ranking on this rerun

Why this is important

The strongest leaderboard model and the safest clinical screening model **can differ**, which is why our framework reports both. In this final rerun, they converged on the same transfer-learning model.



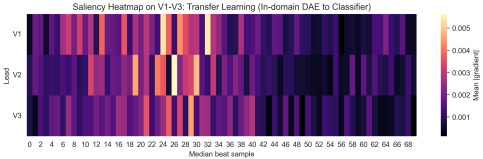
Why This Brings Hope to Healthcare

Hope comes from practicality

- It uses standard 12-lead ECGs already available in real care settings.
- It supports earlier screening attention, where one extra review can matter.
- It still highlights clinically plausible precordial regions, while also using broader cross-lead cues instead of acting like a pure black box.

Hope comes from honesty

- We explicitly surface false negatives.
- We separate aggregate performance from safety-oriented decisions.
- We frame the model as a second reader, not autonomous diagnosis.



Hope is not replacing clinicians.

It is helping them miss fewer dangerous ECGs.

Closing

What we achieved

We built a safety-aware ECG benchmarking pipeline that turns raw 12-lead signals into a practical and clinically grounded Brugada screening aid.

Near-term value

- Triage support
- Second-reader assistance
- Better prioritization for specialist review

Next step

- External validation
- Prospective threshold calibration
- Real clinical workflow integration

From ordinary ECGs to earlier action.

That is the kind of hope AI should bring to healthcare.

Project repository: `https://github.com/avezoor/IDSC2026`